Aalto-yliopisto
Aalto-universitetet
Aalto University

Research Software Engineers

RSEs for CHEM

Aalto University · 26.11.2026

Common pitfalls of research

1. Too many hats

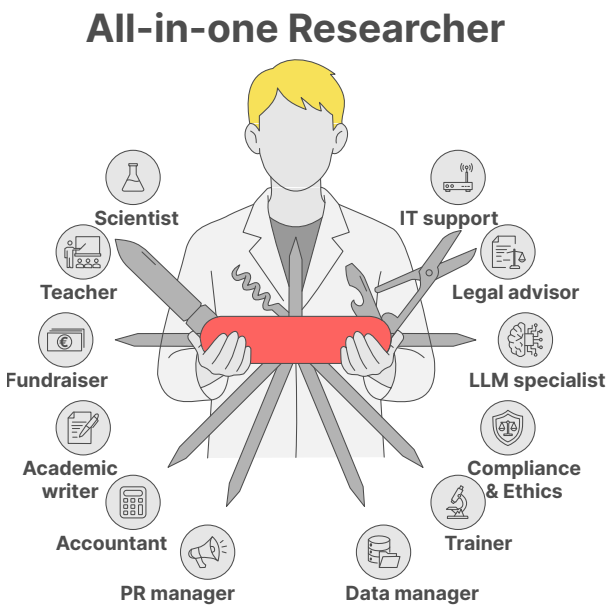
We hire people to do science, but expect them to be their own IT support, data manager, and AI expert.

2. Two Labs, same headache

Two groups a floor apart can lose weeks stuck on the exact same problem.

3. Disconnected starts

A student / PhD / Postdoc takes long to train and leaves data and code behind nobody else understands.



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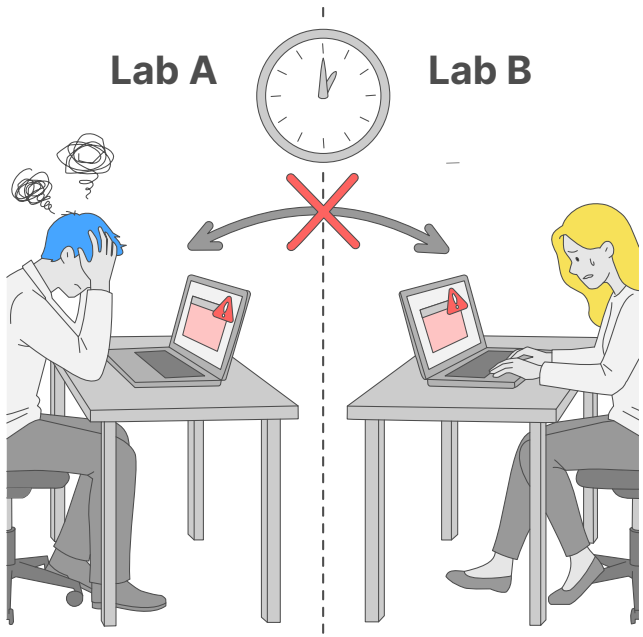
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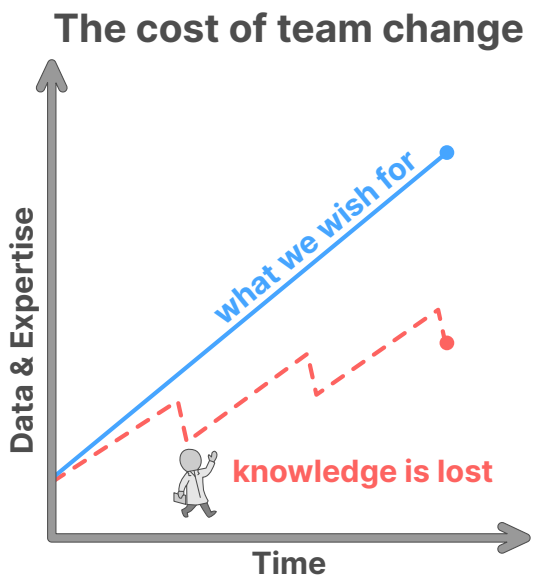
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We make research faster and better

There is too much information for one person to master all of computing and lead their own research.

We take care of the software engineering so researchers can stay focused on their science.

5× time saved

Researchers report a 5× time benefit, comparing the hours we spend to the academic time they save.

01 Avoid struggling with solved problems

Because we work across departments, we spot common roadblocks early and share working fixes to your group.

02 Code that others can run

We help turn fragile, one-off scripts into clean, shareable tools that collaborators and new students can use on day one.

03 Bring old scripts back to present

When an important codebase breaks after someone leaves, we help untangle, update, and rescue it for new projects.

04 Smooth handovers when people change

Better workflows and documentation mean outgoing personnel offboard cleanly, and new hires catch up in days, not months.

How do we do it?

How we help	Programming 50%	Workflows & HPC 35%	Data & Sharing 15%
Do it for you 60%	Build a custom data collection platform for an upcoming study.	Automate your entire analysis pipeline across all input files.	Clean, anonymize, and prepare data for an open data release.
Do it with you 20%	Package your code and publish it as a reusable Python library.	Speed up slow simulation or analysis code so it finishes before the deadline.	Set up a structured database and connect your scripts to it.
Guide you 20%	Fresh eyes to help untangle and fix a stubborn bug.	Show you how to run parallel jobs on Triton or LUMI GPUs.	Advise on fast, reliable file formats for heavy datasets.

85% base-funded Daily drop-in Garage (**15%**) and days-to-weeks projects (**70%**) are free to labs via Aalto base funding. Only long-term projects (**15%**) are billed to grants. **scicomp.aalto.fi/rse**

Start using our services today

CULTURE & LITERACY

Hands-On Training at Scale

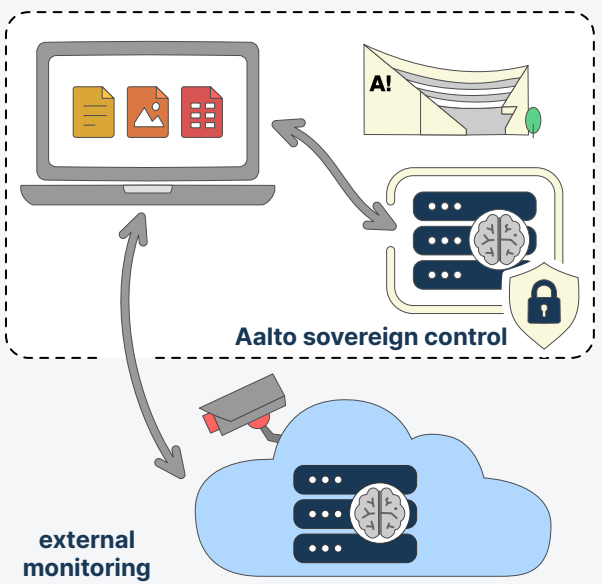
Livestreamed, open-access workshops raising baseline competence across all schools.

- HPC **HPC Kickstart** Triton cluster onboarding & Slurm batch scaling.
- GIT **CodeRefinery** Modular, tested, and reproducible research code.
- ML **Practical Deep Learning** PyTorch GPU profiling & distributed training.
- FAIR **Data Releases** Curated metadata & citable Zenodo DOIs.

SECURITY & INFRASTRUCTURE

Aalto LLM Gateway

llm-gateway.aalto.fi



- **Zero IP leakage**
  Unpublished work never trains external models.
- **AI agent tooling**
  Safe API routing for terminal agents, local models, and automated pipelines (moat).
- **Institutional quota**
  Ensure fair, accountable, and cost-controlled usage of LLMs across the university.

How do we operate?



TIER 1 / DAILY 13:00

SciComp Garage

Low-threshold Zoom & chat help every day at 13:00. Zero cost.

TIER 2 / DAYS-WEEKS

Short projects

Start via ticket or Garage. Hands-on coding & scaling at no cost to you.

TIER 3 / >1 MONTH

Long projects

Dedicated RSE time for major work, with budget agreement.

TIER 4 / PRE-AWARD

Grant co-applying

Partner with us when applying for grants to scope & budget RSE support.

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Who we are: Research Software Engineers (RSEs)

01 Scientists who specialize in software

We have **research** backgrounds ourselves, but our job is building reliable code, data pipelines, and tools.

02 Part of Science-IT (Aalto Scientific Computing)

Science-IT provides the heavy computing infrastructure (Triton) for the whole university. The RSE team is the hands-on human support inside it.

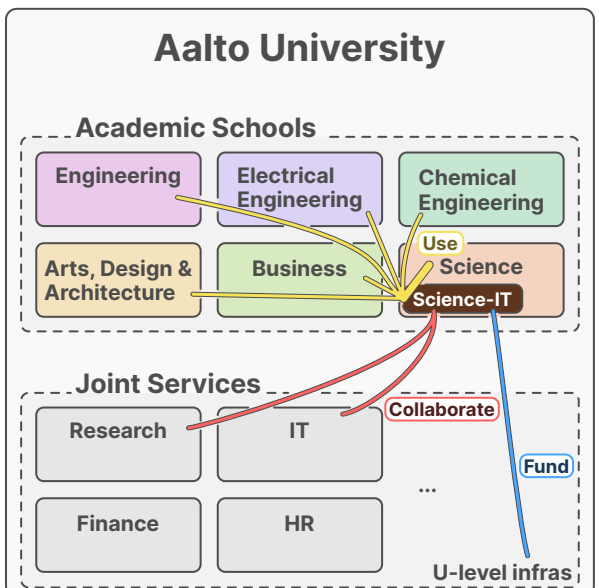
03 Bridging researchers and university services

Hosted in the School of Science, we serve every School and work closely with IT and Research Services.

Come talk to us any workday at 13:00

Drop into the daily SciComp Garage on Zoom.

scicomp.aalto.fi/rse



How can we work together?



Kiitos!

Marijn van Vliet · Patricia Hernandez Leon ·
Pedro Santos Silva · Aalto University

rse@aalto.fi · scicomp.aalto.fi/help/garage/